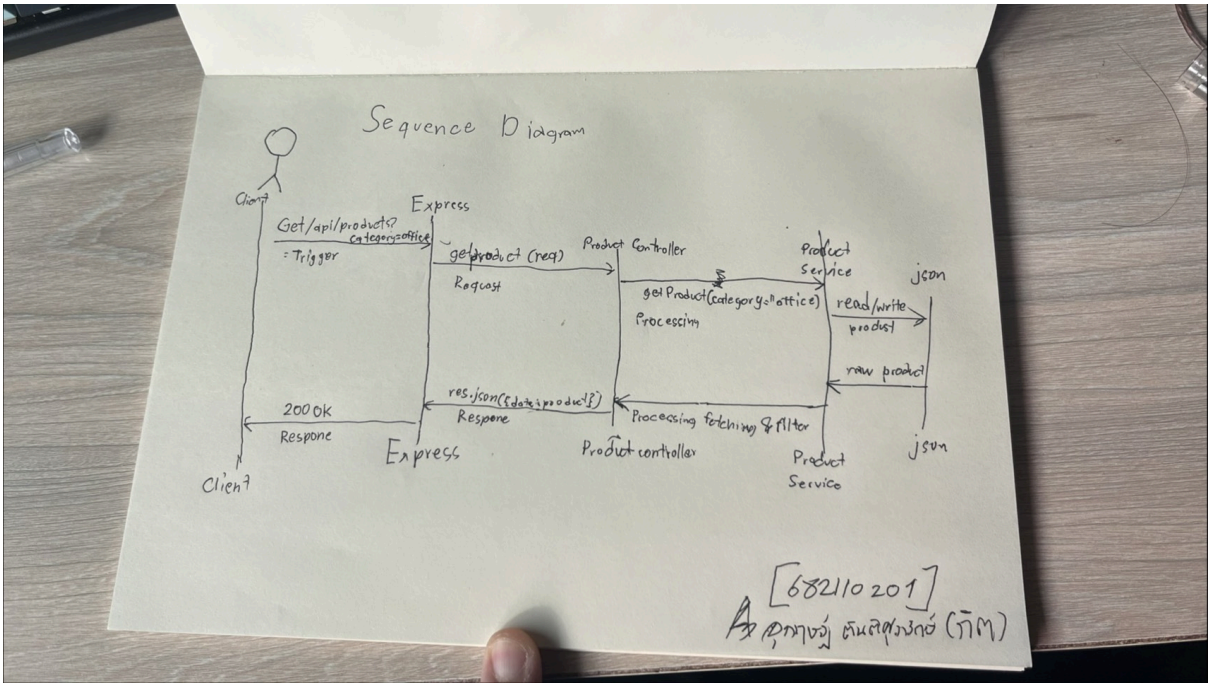
Contract Table: GET /api/products

Field	Type	Required	Description
GET /api/products	HTTP GET	Yes	Fetches the product list
category	string	No	Optional query parameter to filter products by category
200 OK	JSON	Yes	Returns { data: [...] } with matching products
500 Internal Server Error	JSON	No	Returned if product data cannot be loaded or parsed

Sequence Diagram



GenAI Prompt
Write an Express.js API endpoint for GET /api/products that:

reads products from products.json accepts an optional query parameter category returns all products when category is not provided returns only products whose category matches the query when provided uses case-insensitive category matching responds with JSON in the format { data: [...] } handles errors by returning 500 with an error message uses a controller and service layer